Exam (4)

(1)

(a) $$\sum_{n=1}^{\infty} \frac{2\sqrt[3]{n}}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{2(n)^{1/3}}{(n)^{1/2}} = \sum_{n=1}^{\infty} \frac{2}{n^{1/2-1/3}}$$

$$= \sum_{n=1}^{\infty} \frac{2}{n^{1/6}} \qquad \frac{1}{2} - \frac{1}{3} = \frac{3-2}{6}$$

P-Series $\quad P = \frac{1}{6} < 1$

$\therefore \sum_{n=1}^{\infty} \frac{2\sqrt[3]{n}}{\sqrt{n}}$ is diverges by p-series.

(b) $$\sum_{n=0}^{\infty} 5 \cdot \frac{4^{1-n}}{3^n} = \sum_{n=0}^{\infty} 5 \frac{4 \cdot 4^{-n}}{3^n}$$

$$= \sum_{n=0}^{\infty} 20 \left(\frac{1}{12}\right)^n$$

$|r| = \left(\frac{1}{12}\right) < 1$

$\therefore \sum_{n=0}^{\infty} 5 \frac{4^{1-n}}{3^n}$ is converges by geometric

(c) $\sum_{n=2}^{\infty} \frac{2n}{\ln(n)}$ ⇒ Compare to $\sum_{n=2}^{\infty} \frac{n}{n^3}$

$= \sum_{n=2}^{\infty} \frac{1}{n^2}$ P-Series

$P = 2 > 1$ converges

$\sum_{n=2}^{\infty} \frac{2n}{\ln(n)} < \sum_{n=2}^{\infty} \frac{n}{n^3}$

$\therefore \sum_{n=2}^{\infty} \frac{2n}{\ln(n)}$ is converges by Comparison test.

(2) $\sum_{n=1}^{\infty} \frac{8}{n^2+3n+2}$

$\Rightarrow \frac{8}{n^2+3n+2} = \frac{8}{(n+1)(n+2)} = \frac{A}{(n+1)} + \frac{B}{(n+2)}$

$\Rightarrow A(n+2) + B(n+1) = 8$

when $n = -1 \Rightarrow$ $A = 8$

$n = -2 \Rightarrow$ $B = -8$

$$\therefore \sum_{n=1}^{\infty} \frac{8}{n^2+3n+2} = \sum_{n=1}^{\infty} \left(\frac{8}{(n+1)} - \frac{8}{(n+2)}\right)$$

$$S = \frac{\text{first term}}{1 - \text{ratio}} = \frac{4/3}{1-(2/3)}$$

$$\therefore \boxed{S = 4}$$

4/3
2/3 (8/3)
2/5 (10/3)
4/15 (12/3)

(3) $\sum_{n=1}^{\infty} n e^{-n^2}$

(a) Integral test :-

(1) is $n e^{-n^2}$ positive on $[1, \infty)$?

$$n e^{-n^2} = \frac{n}{e^{n^2}} = \frac{\text{positive}}{\text{positive}} = \text{positive}$$

yes, it is positive

(2) is $n e^{-n^2}$ continuous on $[1, \infty)$?

Let $f(x) = \frac{x}{e^{x^2}}$ and $f'(x) = \frac{e^{x^2} - x e^{x^2}(2x)}{e^{x^4}}$

$$\Rightarrow f'(x) = \frac{e^{x^2} - e^{x^2} 2x^2}{e^{x^4}} = \frac{1-2x^2}{e^{x^2}} \neq 0 \text{ for all } "x"$$

$\therefore$ Yes, it is continuous.

③ is ne^{-n^2} is decreasing on $[1,\infty)$?

$\therefore \quad \frac{n}{e^{n^2}} > \frac{n+1}{e^{(n+1)^2}}$ $\qquad \frac{n+1}{e^{(n+1)^2}} = \frac{n+1}{e^{n^2+2n+1}}$

$(n > n+1)$ yes, it is decreasing.

Let $\int_1^\infty \frac{x}{e^{x^2}}\,dx \Rightarrow$ Let $u = x^2$, $du = 2x\,dx$

$\frac{du}{2} = x\,dx$

$$\Rightarrow \int \frac{du}{2e^u} = \frac{1}{2}\int \frac{1}{e^u}\,du$$

$$= \frac{1}{2}\,\frac{e^{-u}}{-1} = -\frac{1}{2e^u}$$

$$\Rightarrow -\frac{1}{2e^{x^2}} \Rightarrow \lim_{x\to\infty} \frac{1}{-2e^{x^2}} = 0$$

$\therefore \sum_{n=1}^{\infty} n e^{-n^2}$ is converges by Integral test.

(b) $S = \frac{\frac{1}{e}}{1-\frac{1}{e}}$

(4) $a_n = \left(1+\frac{2}{n}\right)^{\frac{n}{3}}$

(a) $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(1+\frac{2}{n}\right)^{n/3} \Rightarrow e^{2/3} \neq 0$

$\therefore$ The $a_n = \left(1+\frac{2}{n}\right)^{n/3}$ sequence is converges.

(b) $\sum_{n=1}^{\infty} \left(1+\frac{2}{n}\right)^{n/3}$

Divergence test :-

$\lim_{n\to\infty} \left(1+\frac{2}{n}\right)^{n/3} \Rightarrow e^{2/3}$

$\therefore$ The series $\sum_{n=1}^{\infty} \left(1+\frac{2}{n}\right)^{n/3}$ is diverges by divergence test.

(5) (a) $\sum_{n=1}^{\infty} \frac{n^2 e^n}{n!}$

Use Ratio test :-

$a_n = \frac{n^2 e^n}{n!}$, $b_n = \frac{(n+1)^2 e^{n+1}}{(n+1)!}$

$$\Rightarrow \lim_{n\to\infty} \frac{a_n}{b_n} \Rightarrow \lim_{n\to\infty} \left(\frac{n^2 e^n}{n!}\right)\left(\frac{(n+1)!}{(n+1)^2 e^{n+1}}\right)$$

$$= \lim_{n\to\infty} \frac{n^2 e^n}{n!} \cdot \frac{(n+1)(n!)}{(n+1)^2 e^n \cdot e}$$

$$= \lim_{n\to\infty} \frac{n^2}{e} = \infty > 1$$

$\therefore \sum_{n=1}^{\infty} \frac{n^2 e^n}{n!}$ is converges by Ratio test

(b) $\sum_{n=1}^{\infty} \frac{(2e^n - 1)^n}{e^{n^2}}$

Use Root test :-

$$\lim_{n\to\infty} \left(\frac{(2e^n-1)^n}{e^{n^2}}\right)^{1/n} = \lim_{n\to\infty} \frac{(2e^n-1)}{e^n}$$

$= \infty$ because numerator is larger than Denominator

then $\sum_{n=1}^{\infty} \frac{(2e^n-1)^n}{e^{n^2}}$ is converges by root test.

(6) $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot n!}{n^n}$

Using Ratio test :-

$$a_n = \frac{(-1)^n \cdot n!}{n^n}, \quad b_n = \frac{(-1)^{n+1} \cdot (n+1)!}{(n+1)^{(n+1)}}$$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{(-1)^n \cdot n!}{n^n} \cdot \frac{(n+1)^{n+1}}{(n+1)!\ (-1)^{n+1}}$$

$$= \lim_{n\to\infty} \frac{(-1)^n \cdot n!}{n^n} \quad \frac{(n+1)\ (n+1)^n}{(n+1)\ (n!)\ (-1)\ (-1)^n}$$

$$= \lim_{n\to\infty} \frac{(n+1)^n}{-(n^n)} = -\infty < 1$$

$\therefore$ The $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot n!}{n^n}$ is diverges by Ratio test

(7) $h = 10ft$, returns = $(2/3)$ of previous height

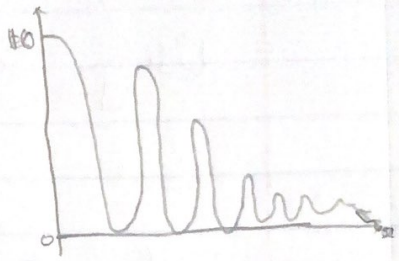
(a)

$$\sum_{n=0}^{25} 10\left(\tfrac{2}{3}\right)^n$$

$$S_{25} = \frac{10}{1-\left(\frac{2}{3}\right)^n}$$

$$\Rightarrow S_{25} = \frac{10}{1-\left(\frac{2}{3}\right)^{25}}$$

$$\Rightarrow S_{25} = \frac{10}{\frac{3^{25}-2^{25}}{(3^{25})}} = \frac{10(3^{25})}{3^{25}-2^{25}} \approx 10$$



(b)

$$\sum_{n=0}^{\infty} 10\left(\tfrac{2}{3}\right)^n$$

$$S_\infty = \frac{10}{1-\left(\frac{2}{3}\right)} = \frac{10}{\frac{3-2}{3}} = \frac{30}{1} = 30\,ft$$